

Week - 01 - Variables, conditions, Loops, Functions\Lecture_2_code.py

```
1  '''
2  Generate a Collatz sequence for a positive integer.
3
4  This solution is not from live coding in the lecture. What we coded in lecture
5  is given below. This solution is a neat version'''
6
7  def generate_collatz_sequence(number: int) -> int:
8      """Print the Collatz sequence and return the number of steps."""
9      steps = 0
10     print(number)
11
12     while number != 1:
13         if number % 2 == 0:
14             number = number // 2
15         else:
16             number = number * 3 + 1
17
18         print(number)
19         steps += 1
20
21     return steps
22
23
24 number = int(input("Enter a positive integer: "))
25 steps = generate_collatz_sequence(number)
26 print()
27 print("Number of steps:", steps)
28
29
30 '''
31 Below this is the code which we wrote in class.
32 We modified it several times to show different aspects, so this code may not
33 make sense to you. But as promised, I upload it
34
35 '''
36 # Input
37
38
39 # Algorithm: Now we put it into a function
40 # Function definition
41 # def gen_collatz_seq(number: int):    # function header / signature
42
43 #     # function body
44 #     while number != 1:
45 #         if number%2:
46 #             number = (number * 3) + 1
47
48 #         else:
```

```
49 #         number = int(number / 2)
50
51
52 #         # Output
53 #         print(number)
54
55
56
57 # input validation
58 # while True:
59 #     number = int(input("Enter a +ive int, (0 to exit): "))
60 #     if number == 0:
61 #         break
62 #     elif number < 0:
63 #         print("You int is -ive, only +ive int are allowed")
64 #         continue
65 #     if number > 0:
66 #         print("You entered correct input")
67 #         break
68
69
70 # if number > 0:
71 #     gen_collatz_seq(number)      # Function call
72
73 # elif number < 1:
74 #     print("You entered a wrong input")
75
76 # else:
77 #     print("Only positive integers are allowed")
78
79
80 # def function_name (What will be the input):
81 #     # algorithm here
82 #     # return some output from this function
83
```